

ABSTRACT

This project investigates whether evolutionary search can be used to design modular robots whose shape, folding behaviour and locomotion are considered together. The chosen platform is the Roblet system, in which small modules join through magnetic connectors, fold from a flat arrangement into a three-dimensional structure, and move under an externally applied magnetic field.

The work developed a graph-based representation of Roblet morphologies and a grammar that limits changes to physically meaningful configurations. Candidate designs were mirrored, checked for collisions and evaluated in MuJoCo. Four objectives were used: folded-gait velocity, change in shape entropy during folding, yaw response to a light stimulus, and speed response to a light stimulus. Attractive and repulsive light responses were treated as separate runs because they require opposite optimisation directions. NSGA-III was used to preserve trade-offs between these objectives, while a proximal policy optimisation (PPO) actor-critic selected grammar actions for mutation and crossover.

The available experiments used a population of 30 individuals, five generations, seven-second rollouts and random seed 42. [The convergence and hypervolume figures indicate measurable progress during the reported run; the exact values and any plateau should be confirmed from the plotted data before submission.](#) The small comparison run also showed that the random baseline achieved a higher best scalarised fitness than the RL-assisted run at this scale. Overall, the project establishes a working physics-in-the-loop framework for studying automated morphology design on a self-assembling robotic platform.

INDIVIDUAL CONTRIBUTION

The physical Roblet platform, its magnetic self-assembly mechanism and the original CAD mass-property data were developed by the Miyashita laboratory and are treated as prior resources in this project [1]. The work reported here consisted of implementing and extending the computational design and evaluation framework. This included the graph genotype and grammar-legal operators, the MuJoCo digital-twin assembly and its MJCF generation, the folded-gait and light-response objective calculations, and the NSGA-III evolutionary loop.

An additional contribution was the adaptation of light-based reaction primitives into attractive and repulsive evolutionary configurations. The PPO actor-critic was implemented to select structural mutation and crossover actions, and a random-action arm was added as a controlled comparison. Hypervolume, convergence, action-distribution and PPO diagnostic plots were also produced. MuJoCo was used for physics simulation, pymoo for multi-objective selection, PyTorch and torch_geometric for the graph policy, and NetworkX for graph manipulation. **[TO COMPLETE: Add the names of lab members who supplied CAD data, starter code or design documents, and confirm the repository or data-access statement required for submission.]**

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Chapter 1

Introduction

1.1 Background and Motivation

1.1.1 Automated and Physics-Aware Robot Co-Design

Imagine asking a robot to design its own body. The fundamental challenge is not simply training a neural network policy to make a fixed physical platform walk; it is discovering the machine’s geometric shape, joint distribution, and dynamic behavior simultaneously. A morphology optimized purely for forward locomotion speed often exhibits poor physical stability. Similarly, a modular body that folds elegantly may fail to transmit magnetic torque effectively, while a configuration that maximizes light-sensor sensitivity might sacrifice movement velocity. In this setting, physical morphology is not a passive container for intelligence, it is an active component of the computational process.

Traditional robotic engineering strictly decouples these design choices into a sequential pipeline. An engineer first specifies the geometric topology, a mechanical modeler tunes joint kinematics, and a control engineer subsequently searches for a control policy that makes the completed platform behave correctly. While effective for highly structured environments, this manual approach restricts exploration to human intuition, often filtering out non-intuitive, high-performing structures before they are ever evaluated in simulation. Automated robot co-design reverses this pipeline: candidate morphologies are generated algorithmically, instantiated in a physics engine, evaluated against task-specific fitness metrics, and returned to the search loop. As il-

illustrated in Figure 1.1, this paradigm merges evolutionary algorithms for macro-level structural search with reinforcement learning to evaluate or guide architectural edits.

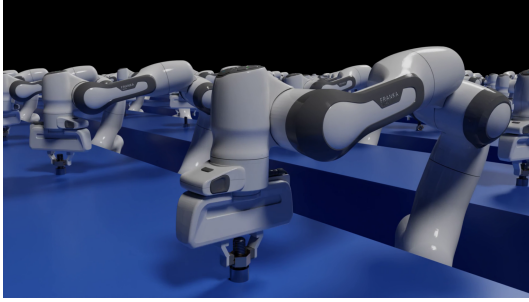
This computational playbook is reflected across key milestones in modern robotics. Large-scale GPU environments such as NVIDIA Isaac Sim (Figure 1.1a) demonstrate how evaluating thousands of parallel physics instances drastically accelerates policy learning. In physical hardware, platforms like Boston Dynamics’ Atlas (Figure 1.1b) highlight how structural dynamics and high-bandwidth control must operate in tight coupling for agile, highly dynamic movement. Crucially, benchmarks like Evolution Gym (Figure 1.1d) establish formal abstractions for morphology-control co-design, decoupling structural generation from behavioral evaluation [2, 3, 4].

The challenge becomes significantly sharper when applied to modular, self-assembling systems. Unlike fixed quadruped or humanoid frames, modular units can combine into an exponential number of topological graphs. For planar systems like Roblets, each graph topology directly determines how 2D flat sheets transform into 3D folded modules (Figure 1.1c), how folded contacts interact with ground friction, and how the physical body responds to external magnetic fields or light signals. Furthermore, local mechanical interlocks and light responses offer a direct pathway toward collective swarm intelligence.

The objective of this dissertation is to bridge these paradigms: uniting evolutionary multi-objective search, deep reinforcement learning structural edit policies, and physics-in-the-loop evaluation inside a modular pipeline where a candidate design begins as an abstract graph, self-folds into a 3D physical body, and is evaluated directly by its embodied physical capabilities.

Image sources and attributions:

- **(a)** Source: Page capture from the official NVIDIA Isaac Sim website, accessed 4 September 2026.
- **(b)** Source: Page capture from the official Boston Dynamics website, accessed 4 September 2026.
- **(c)** Source: Page capture from the official Stanford Robotics website, accessed 4 September 2026.



(a) Parallel physics simulation (NVIDIA)

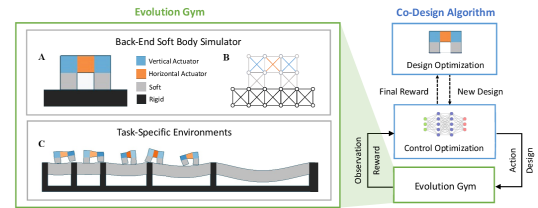


(b) Embodied dynamic platform (Atlas)



(c) Planar 2D-to-3D self-folding mechanism

3 Evolution Gym



(d) Morphology-control co-design loop

Figure 1.1: Key paradigms informing automated modular robot design: (a) large-scale parallelized physics simulation on GPUs; (b) dynamic physical hardware requiring tight body-brain integration; (c) planar sheet transformations into 3D folded morphologies via thermal hinges; and (d) computational co-design loops combining evolutionary search with policy learning [2, 3, 4].

- (d) Source: Page capture from the official MIT CSAIL website, accessed 4 September 2026.

1.1.2 The Roblet Platform

Roblets are small, modular units with magnetic bonding sites and foldable sections. The units can be assembled from a two-dimensional arrangement and then folded into a three-dimensional robot. Locomotion is produced by an externally applied oscillating magnetic field rather than by motors placed in every module. This combination makes Roblets a useful test case for automated design because morphology affects assembly, folding, magnetic response and motion at the same time [1].

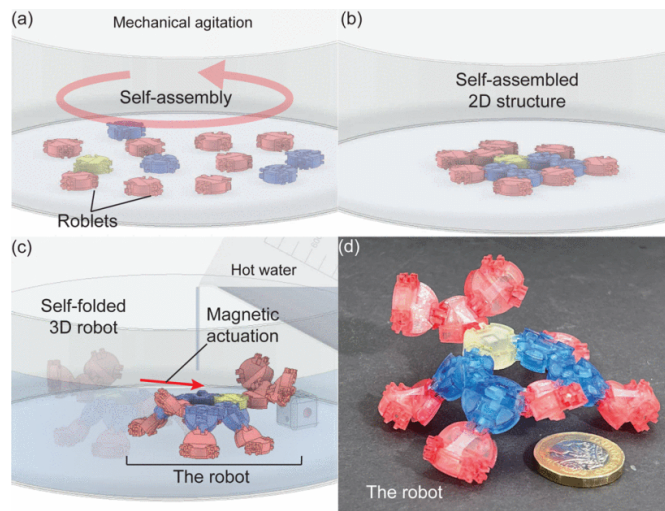


Figure 1.2: Roblet assembly, folding and actuation. (a) Magnetic self-assembly of modules into a 2D configuration. (b) Heat-triggered self-folding into a 3D structure. (c) Locomotion under an oscillating magnetic field. Source: Han et al., IROS 2023 [1].

1.1.3 Swarm Communication via Light-Based Virtual Pheromones

Light-sensitive joints provide a simple local reaction primitive. A joint can respond to a one-sided light stimulus by changing the robot's turning behaviour, or to a wider stimulus by changing its speed. These responses are relevant to virtual pheromone communication because a robot could, in principle, move towards or away from a signal produced by another robot. In this project the primitive was evaluated on one simulated robot at a time. No multi-robot swarm experiment was carried out, so the results concern

a building block of swarm behaviour rather than swarm coordination itself.

1.2 Problem Statement

Existing Roblet morphologies are designed by hand, which limits the range of structures that can be considered [1]. The difficulty is increased because a candidate is defined by interacting module types, port connections, module count, fold angles and light-response settings. Even before all graph connections are counted, allowing between 2 and 40 modules creates a large discrete search space.

The problem addressed in this report is the limited integration of morphology generation, self-folding behaviour and locomotion performance in an evolutionary robotic system. The central hypothesis is that an ML-assisted evolutionary framework can generate legal two-dimensional Roblet configurations, evaluate their folded three-dimensional behaviour and use the resulting measurements to improve the population. A second question is whether entropy-based folding complexity and light-response measures can reveal relationships between morphology and behaviour. The random-baseline experiment is used later to assess the selected ML mechanism; it is not the main hypothesis.

1.3 Aims and Objectives

The project aim was to develop an ML-assisted evolutionary framework within the Roblet platform for designing two-dimensional module morphologies that self-assemble, self-fold and locomote magnetically. The basic objectives were to: (O-1) develop a Python evolutionary approach for generating self-assembling configurations; (O-2) extend the simulation to support two-dimensional to three-dimensional self-folding; (O-3) evaluate folded structures under magnetic actuation and feed their fitness back into evolution; and (O-4) analyse evolved morphologies and their influence on locomotion across generations.

The original advanced objective AO-1 concerned comparing surrogate-learning methods for fitness prediction. It was changed during the project to the development of attractive and repulsive light-response objectives, since direct parallel simulation and caching made surrogate prediction less necessary and the reaction primitives were more

closely related to the platform motivation. AO-2 remained the investigation of entropy and morphology complexity in relation to locomotion and evolutionary outcomes.

1.4 Contributions

The main contributions are as follows:

- A graph-based Roblet genotype with symmetry construction and grammar-legal structural edits.
- A MuJoCo evaluation path from a two-dimensional graph to a folded, magnetically actuated three-dimensional assembly.
- A four-objective formulation combining locomotion, folding complexity and light-response behaviour, with separate attractive and repulsive runs.
- A PPO graph policy that selects mutation and crossover actions, together with a uniform-random comparison arm.
- Evaluation tools for convergence, hypervolume, Pareto structure, action use and RL training behaviour.

The intended contribution is the combination of policy-guided graph editing for both mutation and crossover with many-objective selection and Roblet-specific constraints, rather than discrete RL actions alone.

1.5 Project Management

The development and execution of this project were tracked against a structured 17-week schedule running from 15 May 2026 to 07 September 2026. Table 1.1 details the planned task timelines alongside actual completion dates and key technical outcomes recorded during development.

Initial development proceeded on schedule, with foundational XML generation and hinge self-folding mechanisms established in early July. However, actual progress diverged from the baseline plan across key technical phases:

Table 1.1: Planned vs. actual project timeline across major engineering and evaluation phases.

Project Phase	Planned Dates	Actual Dates	Key Technical Outcomes
Literature Review & Back-ground	15 May – 12 Jun	15 May – 27 Jul	Initial background review; extended through mid-July during formal review submission.
Simulation Framework & Physics Engine	12 Jun – 26 Jun	02 Jul – 25 Jul	Self-folding XML setup, joint torque control, and oscillating magnetic actuation.
Morphology Representa-tion	26 Jun – 10 Jul	13 Jul – 01 Aug	Directed graph serialization, visualizer in-terface, and port connection logic.
Evolutionary Framework Integration	10 Jul – 24 Jul	02 Aug – 23 Aug	EA pipeline initialization, grammar-masking collision gates, symmetry en-forcement, and shape entropy.
ML-Assisted Optimisation (PPO / RL)	07 Aug – 14 Aug	24 Aug – 02 Sep	Pheromone response metrics, RL actor-critic integration, and action-penalty tun-ing.
Simulation Experiments & Analysis	14 Aug – 28 Aug	12 Aug – 31 Aug	Headless parallel simulations, magnetic sweep optimization, and convergence plotting.
Dissertation Writing & Re-vision	14 Aug – 04 Sep	26 Aug – 04 Sep	Final thesis compilation, manuscript drafting, and algorithmic bug fixes.

Simulation Framework Bottlenecks and Downstream Schedule Shifts: Building the physics simulation environment and dynamic assembly generator consumed nearly four weeks of intensive development (from early July through late July), significantly longer than the planned two-week allocation. Debugging low-level physical dynamics—such as hinge joint flapping, accurate module spatial layout in XML, connector inertial mass adjustments for embedded magnets, dynamic torque actuation under oscillating magnetic fields, and fluid/glass interactions, required extensive iteration across dozens of code refinements. Because multi-objective optimization (MOO) and reinforcement learning (RL) both rely on a stable, bug-free physics evaluator, downstream framework integration was intentionally held back until high-fidelity simulation and headless parallel execution were completely validated in mid-August.

Pivot from ML Surrogate Fitness to Direct Pheromone Metrics (AO-1 Adjustment): The original project objectives specified training a machine learning surrogate model to predict physical locomotion fitness to accelerate evolutionary search. During mid-August, when headless parallel MuJoCo simulations achieved high throughput (around 1.2 minutes per generation), surrogate fitness prediction was deemed unnecessary. Instead, the advanced objective was pivoted toward implementing direct light-response and virtual pheromone attraction/repulsion behavior metrics. This modification aligned the physics-in-the-loop pipeline with high-fidelity behavioral evaluation without relying on surrogate approximations.

Delayed RL Integration and Redesign Phase: While evolutionary pipeline integration was scheduled for late July, establishing stable multi-objective constraints (collision checks, mirror symmetry, and shape entropy) extended through mid-August. Consequently, the implementation of the PPO actor-critic network for RL-guided graph edits commenced later than planned, in late August. Initial diagnostic runs revealed policy collapse caused by excessively severe penalties on invalid grammar actions and elitism handling issues in NSGA-III. Reallocating contingency time in late August enabled targeted redesign of penalty structures and operator selection routines, successfully stabilizing the actor-critic policy before final thesis compilation.

1.6 Dissertation Structure

Chapter 3 introduces the hardware and explains how its main physical effects were represented in MuJoCo. Chapter 4 gives the design of the evolutionary framework, beginning with the genotype and ending with the complete computational pipeline. Chapter 5 states the settings used for the reported experiments and defines the controlled random baseline and evaluation metrics. Chapter 6 presents convergence, trade-offs, morphology, baseline comparison and RL diagnostics, then relates the findings to the objectives. Chapter 7 closes with the delivered contributions, limitations and future experiments.

Chapter 2

Literature Review

Designing autonomous modular robots requires addressing both physical mechanics and algorithmic optimization. Historically, creating functional robot morphologies depended heavily on human engineering intuition, a process that inherently restricts exploration to familiar configurations. To overcome this, the automated design of self-assembling and self-folding modular robots demands a synthesis of physical hardware platforms, evolutionary multi-objective optimization, graph-structured representations, and reinforcement learning.

This literature review evaluates the foundational frameworks and recent advances across five key domain areas. Section 2.1 surveys the landscape of modular and self-reconfigurable robots, positioning the unique physical traits of Roblets relative to existing hardware systems. Section 2.2 examines evolutionary robotics, focusing on structural representation and search-space management. Section 2.3 critically evaluates reinforcement learning techniques applied to structural search versus control optimization. Section 2.4 analyzes multi-objective evolutionary algorithms, focusing on high-dimensional Pareto optimization. Section 2.5 reviews graph-structured representations and graph neural network policies for variable-morphology systems. Finally, Section 2.6 synthesizes these findings to define the precise research gap addressed in this work.

2.1 Modular and Self-Reconfigurable Robots

Modular robotic systems build functional machines by combining standardized physical building blocks. The foundational principles of programmable self-assembly estab-

lished by Klavins (2007) demonstrated that complex physical macro-structures could emerge from simple units governed by local stochastic interactions and discrete bonding rules [5]. Early modular architectures, however, relied primarily on rigid mechanical latches and required structured, high-energy environments to successfully assemble.

Over the last two decades, hardware platforms have progressed significantly toward parallelized self-assembly and higher operational autonomy. Systems such as SMORES-EP (Liu et al., 2023) achieve versatile reconfiguration by leveraging electropermanent magnets, enabling low-power dynamic coupling [6]. Similarly, recent advancements in spherical freeform modular robots (Tu et al., 2026) demonstrate high degrees of self-reconfiguration autonomy and flexible locomotion across unstructured terrain [7]. Taking the concept of physical adaptation further, Wyder et al. (2025) proposed the concept of "robot metabolism," where machines dynamically adapt by consuming and reallocating structural components from other modular units [8]. While these platforms illustrate significant mechanical flexibility, comprehensive surveys of the field (Alattas et al., 2019) highlight a recurring bottleneck: as the number of available physical interfaces scales, hand-crafting effective topologies quickly becomes intractable due to the massive combinatorial search space [9].

The Roblet platform (Han et al., 2023) represents a distinct paradigm within modular robotics. A Roblet begins as a flat, 2D 3D-printed unit equipped with magnetic "jigsaw smart-glue" passive bonding interfaces. When exposed to thermal activation (e.g., hot water), heat-responsive PVC hinges trigger precise self-folding mechanisms that transform the 2D sheet into a 3D functional module. Once folded, locomotion is driven externally through oscillating magnetic fields, while light-based "virtual pheromones" govern collective swarm behavior without requiring bulky onboard motors or battery packs.

However, despite these innovative mechanics, existing Roblet configurations reported in prior art (Han et al., 2023) remain strictly hand-designed. Human intuition struggles to predict how complex 2D planar layouts will fold into 3D multi-hinge structures, or how those 3D shapes will interact with oscillating magnetic fields to produce stable gaits. To unlock the potential of the Roblet platform, an automated, computational co-design approach is required.

2.2 Evolutionary Robotics and Morphology-Control Co-Design

Evolutionary Robotics (ER) applies the principles of natural selection to simultaneously synthesize a robot’s physical morphology (body) and controller (brain). A broad survey of modular design automation (Li et al., 2023) categorizes the field into three core challenges: representation (genotype-to-phenotype mapping), search space exploration, and controller-morphology co-optimization [3].

A central challenge in evolving modular structures is managing a large, constrained design space. Work on evolutionary robot design also considers interactions among design variables, including epistatic effects, while Evolution Gym highlights the importance of expressive design representations and the difficulty of co-optimizing body structure and control as the search space grows [10, 2]. In this project, grammar masking is introduced as a method contribution: it enforces topological and kinematic constraints, such as valid port adjacency, before a candidate is sent to physics simulation. This implementation choice is intended to reduce evaluations of plainly illegal designs; it is not presented as an established conclusion of the cited literature.

In terms of structural representation, graph-based heuristic search methods (Xu et al., 2021) provide one example of modeling robot topologies for design search [11]. Representing robot bodies as directed graphs allows evolutionary operators to operate on discrete subtrees. However, evaluating candidate morphologies presents a computational bottleneck. Benchmark platforms such as Evolution Gym (Bhatia et al., 2021) provide standardized grid/voxel-based soft-robot environments for large-scale evaluation [2]. While useful for general algorithm comparison, generic voxel benchmarks do not capture all of the specific physical constraints of Roblets, such as planar 2D-to-3D self-folding kinematics, magnetic field actuation, and directional light-sensing responses. Consequently, a dedicated, physics-in-the-loop simulation environment grounded in MuJoCo is necessary to evaluate Roblet candidate designs against real-world physical dynamics.

2.3 Reinforcement Learning for Robot Design

Reinforcement Learning (RL) is widely used to learn control policies for existing robot bodies, including in micro- and nanorobotics [4]. By contrast, morphology-control co-design remains a computationally demanding problem: surveys and co-design studies identify the interaction between body and controller search as a central challenge, while Evolution Gym notes that combining design optimization with deep RL remains difficult in harder environments [3, 12, 2].

Applying RL to structural design can therefore involve discrete design decisions in addition to continuous control. The cited literature includes RL-based morphology decisions and joint morphology-control optimization, but it does not establish PPO graph-edit actions as a standard alternative to other EA-RL hybrids [13, 12, 14]. The present thesis uses PPO to select structural actions within an evolutionary loop; this is the design being evaluated here, rather than a capability claimed for the cited studies.

Hybrid frameworks combining Evolutionary Algorithms (EAs) and RL are consequently relevant to this problem. Across the cited work, EAs commonly provide population-based morphology search while RL learns or refines behaviour, although the division of labour varies between methods [3, 12, 14]. Other work uses RL-inspired signals to inform evolutionary operator choices, illustrating that hybridisation need not mean that an RL policy directly performs every structural edit [15].

The approach developed in this thesis uses a Proximal Policy Optimization (PPO) agent to select structural actions, such as node additions, subtree prunings, and crossover subtree swaps, on candidate graph genotypes inside an active evolutionary loop. Within the supplied corpus, this appears distinct from the cited co-design baselines because those sources do not directly test PPO over graph mutations and crossover. The distinction is therefore a positioning of the present method, not a claim that all prior body-brain co-design methods restrict RL to motor control.

2.4 Multi-Objective Evolutionary Algorithms in Robotics

Real-world robot design inherently involves balancing conflicting performance requirements. A morphology optimized purely for forward locomotion speed often exhibits poor stability or excessive structural complexity. Multi-Objective Evolutionary Algo-

gorithms (MOEAs) address this by discovering a Pareto front, a set of non-dominated solutions representing optimal trade-offs between competing criteria.

Standard multi-objective algorithms, such as NSGA-II, rely on crowding distance metrics to maintain population diversity along the Pareto front. However, when the optimization problem involves more than three objectives (many-objective optimization), crowding distance metrics lose selectivity because almost all individuals become non-dominated relative to one another. To resolve this, Deb and Jain (2014) developed NSGA-III, which replaces crowding-distance sorting with reference-direction-based niching. By projecting candidate solutions onto uniformly distributed reference lines in objective space, NSGA-III maintains high population diversity and convergence pressure across higher-dimensional trade-off surfaces.

In robotic co-design, integrating machine learning with MOEAs is relevant when the design space is constrained. Ming et al. (2024) provides a related example of deep reinforcement learning selecting genetic operators in constrained multi-objective optimization [16]. This project applies NSGA-III to a four-objective optimization landscape:

1. Maximize folded-gait locomotion velocity (f_1),
2. Maximize entropy-based folding complexity gain ($\Delta H = H_{3D} - H_{2D}$) (f_2),
3. Maximize/minimize light-based pheromone yaw response (f_3),
4. Maximize/minimize light-based pheromone speed response (f_4).

NSGA-III is well-justified for this many-objective setting because its reference-direction niching is designed to maintain diversity and selection pressure when several objectives make ordinary crowding-based selection less discriminating [17].

2.5 Graph-Structured Policies for Variable Morphology

A persistent challenge in evolving variable modular architectures is learning across candidates that differ in node counts, degrees of freedom, and topological layouts. Variable morphologies create changing dynamics and state-action distributions, which complicate shared learning across designs [12, 3].

Graph-based policies appear promising for variable-morphology settings because they can represent relationships between modules and connections directly. Existing work in the supplied corpus also uses graph-based search for robot design and morphology-aware or morphology-agnostic policy learning [11, 18, 19, 20]. These sources do not establish a single standard graph-policy architecture or directly document the particular attention and operator details used in this thesis.

These studies motivate morphology-aware representations, while leaving open how a policy should select legal graph edits within an evolutionary search loop. This thesis addresses that narrower design question with a graph encoder and PPO policy. The broader literature also includes studies of adaptive or life-like behaviour in flexible and active structures, which reinforces the value of evaluating morphology together with behaviour rather than treating shape as an isolated variable [21].

This work makes two implementation choices beyond the cited representations:

1. **Per-Edge Multi-Head Attention:** A dedicated Graph Transformer encoder with multi-head attention per edge is used to capture directional flow and structural dependencies across Roblet connection ports.
2. **Unified Structural Edit Policy:** Rather than using graph policies strictly to produce motor control torques, the Continuously-trained PPO agent uses the Graph Transformer encoder to directly output probability distributions over discrete structural edits (mutation and crossover subtree operations) across arbitrarily sized modular genotypes.

2.6 Summary and Research Gap

While substantial progress has been achieved in modular robotics, evolutionary algorithms, and reinforcement learning, existing methodologies remain fragmented across these domains:

- Modular platforms like Roblets (Han et al., 2023) offer compelling mechanical properties, such as heat-driven 2D-to-3D self-folding and external magnetic actuation, but their design space remains limited by manual human engineering.

- The supplied co-design literature supports a distinction between RL for control and evolutionary search over design, but does not establish a direct robot-morphology example of PPO choosing graph-edit mutations and crossover.
- The cited sources motivate combining learned decision-making with evolutionary and many-objective search, while the particular integration of PPO-selected graph edits, NSGA-III and Roblet-specific constraints is the method evaluated in this thesis.

Research Gap Statement: Within the supplied corpus, no directly comparable study is identified that combines (a) policy-guided evolutionary graph editing for both structural mutation and crossover, (b) high-dimensional (4-objective) NSGA-III Pareto search, and (c) domain-specific kinematic, folding, and light-response grammar constraints within a unified, physics-in-the-loop co-design pipeline. This corpus-bounded gap motivates the method developed in the following chapters; it is not an exhaustive claim about all prior work.

The following chapters detail the methodology developed to address this gap, presenting an automated evolutionary framework capable of discovering functional, highly non-intuitive Roblet morphologies.

Chapter 3

The Roblet Platform and Physics Simulation

3.1 Physical Roblet Hardware



(a) Top view.



(b) Side view.

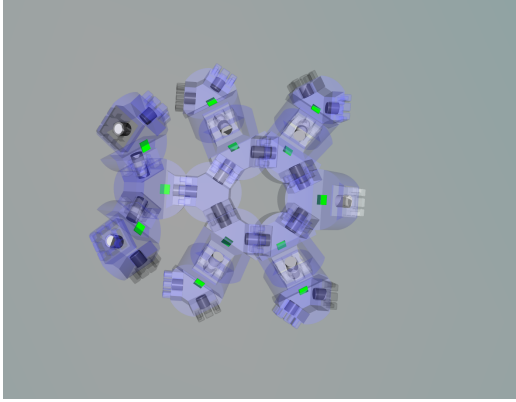
Figure 3.1: Physical Roblet modules under the microscope/camera setup used for hardware validation, showing the translucent 3D-printed module bodies and embedded permanent magnets (red markers) at each port. Photograph of the physical platform developed by the Miyashita Lab [1]; shown here for direct comparison with the simulated digital twin (Fig. 3.2).

3.2 Actuation and Self-Folding Principles

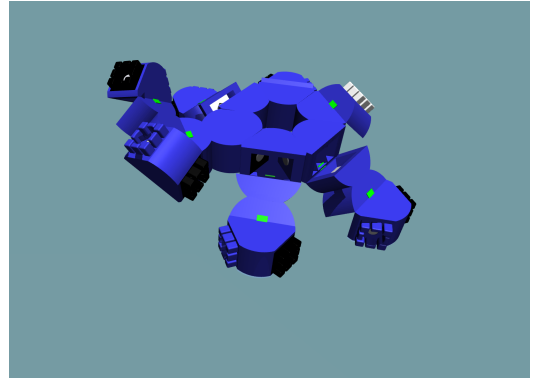
The physical design uses heat-responsive PVC hinges to convert the flat arrangement into a folded structure. In the simulation, this process is represented by allowing connected module bodies to settle into their folded configuration rather than prescribing the final pose directly. The assembled robot is then driven by an oscillating magnetic field. Each embedded magnet experiences a torque that depends on the field direction, and the combined torques produce the motion of the whole assembly.

The light-response mechanism is represented as a change at selected foldable joints. A one-sided stimulus is used to measure yaw response, while a wider stimulus is used to measure speed response. These two measurements form the reaction primitives used later in the evolutionary objective set. The simulator therefore provides a common route from morphology to folding, actuation and measurable behaviour.

3.3 MuJoCo Digital-Twin Simulation



(a) Top view, transparent bodies showing embedded magnets (green)



(b) Rendered assembly

Figure 3.2: Simulated Roblet digital twin in MuJoCo, corresponding to the physical assembly in Fig. 3.1. Panel (a) shows the transparent top-down view used to verify embedded-magnet placement against the CAD source data; panel (b) shows the fully rendered simulated assembly.

The digital twin was implemented as a set of independent module bodies with mass, centre-of-mass and inertia values taken from the available CAD exports. A cylindrical

N42SH magnet model with magnetic moment $m = 6.5 \times 10^{-3} \text{ A m}^2$ was placed at each connector. Connected bodies were joined with weld equality constraints. This choice allows the bodies to remain separate physical objects while their assembled relationship is enforced during settling; it is closer to the assembly process than a purely scripted kinematic tree.

The external field applies the torque $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$ to each magnet. For locomotion, candidate field strengths of 0.001, 0.008 and 0.05 T were tested. The field was swept for each morphology because a small or heavy structure may stall at a field that drives another structure into excessive rolling or tumbling. The candidate with the highest stable velocity was retained. [TO COMPLETE: Confirm the source of the CAD mass properties and add the exact settling and rollout durations if they differ from the current code settings.]

Chapter 4

Evolutionary Framework Design

4.1 Genotype Representation and Design Variables

Each candidate is represented as a rooted directed graph. Nodes represent modules and edges represent port-to-port connections. A node can be rigid, mountain-fold or valley-fold. Port 1 is used for the parent connection, while ports 2 and 3 provide possible child connections. The full morphology contains between 2 and 40 modules after mirroring.

The design variables include module type, port adjacency, module count, ordinary hinge angle, light-sensitive joint selection and light-detection hinge angle. The hinge angle controls the folded geometry, while the light angle controls the response produced when a selected joint receives light. Keeping these variables in one genotype allows the search to evaluate their interaction rather than optimising shape and response separately.

4.1.1 Bilateral Symmetry and Mirror Construction

Only one half of the morphology is evolved directly. The root module and its spine child provide the mirror anchors, while the expandable side is connected through the remaining port. During construction, the other side is generated deterministically by reflecting the evolved half. This keeps the final robot bilaterally symmetric and reduces the number of graph decisions required by the search.

The mirror is applied at build time rather than treated as a separate learned action. As a result, every mutation and crossover operates on the same half-graph representation, while the simulator receives the complete structure. The representation reduces

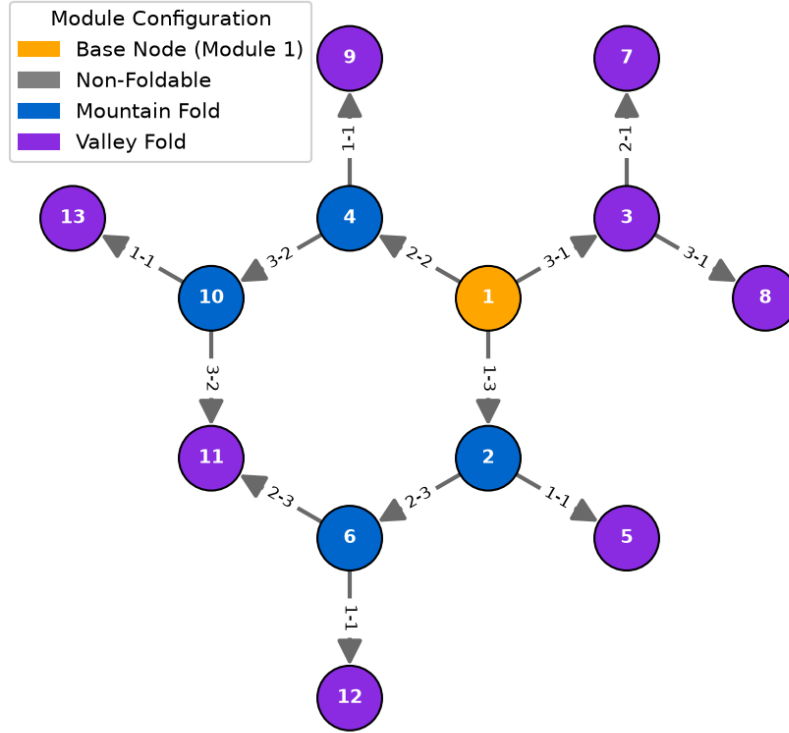


Figure 4.1: Example half-genotype as a directed graph: nodes are modules (coloured by fold type, orange = root module 1), edges are labelled by the (source port)-(destination port) connector pair. Only this half-graph is evolved directly; the full bilaterally symmetric morphology is produced by mirroring it about the root’s spine (Section 4.1.1).

redundant solutions without removing the structural edits available to the evolutionary operators.

4.1.2 Grammar-Legal Mutation and Crossover Operators

The operator set contains eight mutations and two crossovers: add node, delete node, prune subtree, change fold type, change hinge angle, reconnect a port, toggle a light sensor, change the light hinge angle, graft a subtree and swap subtrees. Each action is masked according to the current graph. Addition requires a free port and the module limit, deletion protects the root and targets a removable leaf, and reconnection must leave valid port assignments.

Grammar masking was selected instead of generating an illegal graph and rejecting it afterwards. Reject-and-retry wastes simulation and policy samples, while a soft penalty asks the policy to learn rules that can be enforced directly. Masking guarantees

that sampled structural edits are legal and leaves simulation failure to represent physical or numerical problems rather than simple grammar errors.

4.2 Multi-Objective Fitness Formulation

$$\mathbf{f} = \left[\underbrace{v_b}_{f_1}, \underbrace{\tilde{H}_{3D} - \tilde{H}_{2D}}_{f_2}, \underbrace{\Delta\psi_L - \Delta\psi_b}_{f_3}, \underbrace{\frac{v_F - v_b}{v_b}}_{f_4} \right] \quad (4.1)$$

Here, v_b is the baseline folded-gait velocity, $\tilde{H}_{2D}$ and $\tilde{H}_{3D}$ are the normalised shape entropies before and after folding, $\Delta\psi_b$ and $\Delta\psi_L$ are the baseline and left-stimulus yaw changes, and v_F is the velocity under the front stimulus. The four components are maximised in the attractive run. In the repulsive run, the signs of the two light-response preferences are reversed by the objective configuration rather than by changing the measured signed quantities. The objective vector is evaluated after folding and actuation. The first component measures folded-gait velocity. The second measures the change in normalised shape entropy between the flat and folded forms. The final two components measure changes in yaw and speed under light stimulation relative to their baselines. All four objectives are maximised in the attractive run; the optimisation direction of the two light-response objectives is changed in the repulsive run.

4.2.1 Folded-Gait Locomotion and Shape-Entropy Objectives

Velocity is measured after the torque ramp so that the initial settling period does not dominate the result. The entropy calculation voxelises the morphology at more than one spatial scale and combines the resulting occupancy variation into a normalised value. The reported folding-complexity objective is $f_2 = \tilde{H}_{3D} - \tilde{H}_{2D}$, so a positive value indicates greater measured structural complexity after folding. The raw two-dimensional and three-dimensional entropy values are retained for the later relationship analysis.

4.2.2 Pheromone Reaction-Primitive Tests

Each light-response objective uses two stages. The first stage measures the baseline behaviour without the relevant light stimulus. The second stage applies either a left-side stimulus for yaw or a front stimulus for speed and measures the changed response.

The yaw objective is the difference between the light and baseline rotations, while the speed objective is the relative change from baseline speed.

A separate baseline stage is important because the robot can naturally curve or drift during its magnetic gait. The comparison must therefore use a travel-aligned local frame established for that objective, rather than the trajectory produced during the stimulated rollout. The settled pose is also not physically rotated simply to define the frame. This keeps the measured difference attributable to the light condition as far as the simulation permits and avoids confusing pre-existing gait direction with reaction behaviour.

4.2.3 Attractive versus Repulsive Response Configuration

Attractive and repulsive behaviour were implemented as separate evolutionary runs. Both runs use the same signed measurements: positive yaw denotes turning towards the light and positive speed response denotes speeding up. The optimisation direction is then changed for the repulsive run so that turning away and slowing down are preferred. A single run containing both directions would ask the same light-hinge variable to increase and decrease at once, creating a self-contradictory objective pair. Separate runs make the intended behaviour explicit and allow their morphologies to be compared directly.

4.3 RL-Guided Actor-Critic for Structural Edits

The structural-edit policy receives an ordered pair of parent half-genotypes. Each graph is represented by nine node features: module type, normalised hinge angle, depth, free-port fraction, root status, light sensitivity and light-hinge angle. A Graph Transformer encoder produces representations for the variable-size graphs, using hidden dimension 32, two Transformer blocks and four attention heads. Separate actor and critic networks use this graph representation and a global mean readout.

The action is sampled hierarchically. First, the actor selects one of eight mutation actions or two crossover actions. It then selects legal target nodes, ports and parameters. The mask is applied at each stage, so the policy cannot select a root deletion, an unavailable port or an action that exceeds the module limit. The reward is the change in scalarised four-objective fitness from parent to child, with a penalty for an unstable or

failed simulation. PPO is updated once per generation using the offspring decisions as one-step transitions.

This policy guides variation, not locomotion control. The policy chooses how the graph is changed, while MuJoCo supplies the physical evaluation. **[TO COMPLETE: Confirm whether the final submitted implementation uses a shared actor-critic encoder and redesigned entropy schedule; the available hyperparameter notes describe separate actor and critic weights and a fixed entropy coefficient.]**

4.4 NSGA-III Environmental Selection

NSGA-III ranks individuals using the complete four-objective vector rather than the scalarised value used for the PPO reward. All objectives are converted to a minimisation form for pymoo, while the attractive or repulsive configuration determines the direction of the two light-response objectives. Reference directions are used to retain solutions across the objective space. This is appropriate for four objectives because a single best score would hide useful trade-offs, while reference-direction niching helps maintain spread.

Feasibility is handled before objective comparison. Structural validity is enforced during graph construction, mirroring and grammar-masked variation, so illegal structural individuals do not need a separate violation score. Simulation failure or instability is treated separately as an infeasible evaluation. Thus, the collision gate handles representation-level feasibility, whereas the simulation constraint records whether a legal structure can also be evaluated successfully.

4.5 Implementation and Computational Pipeline

The implementation evaluates individuals in parallel operating-system processes. For each generation, graph data are converted to MJCF files, checked and passed to MuJoCo workers. Results are stored with the individual and are used by both the PPO update and NSGA-III selection. Checkpoints contain the generation, population, random-number state, training state and objective-normalisation information, allowing a run to resume without restarting from the initial population.

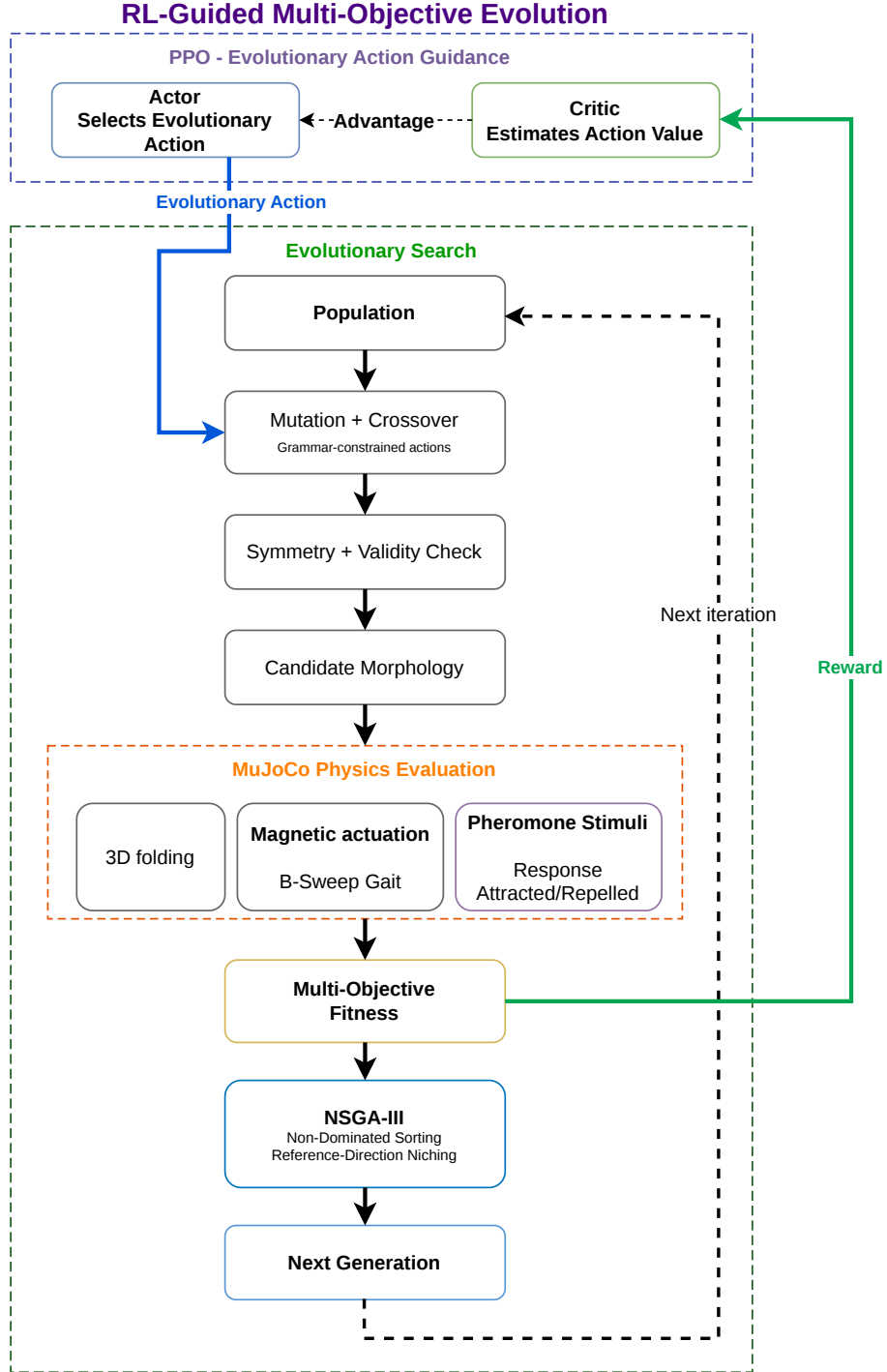


Figure 4.2: End-to-end evolutionary pipeline. The PPO actor selects a structural edit (mutation or crossover) from the population pool, which is grammar-masked, mirrored for bilateral symmetry, and passed through a 3D collision gate before parallel MuJoCo evaluation (magnetic B-sweep gait, then the pheromone reaction-primitive tests). Resulting objectives feed both the PPO reward/advantage signal and NSGA-III’s reference-direction non-dominated sorting, which determines the next generation.

An evaluation cache hashes deterministic graph evaluations and avoids simulating unchanged individuals again, including carried-over survivors. The project summary reports approximately 1.2 minutes per generation with parallel evaluation. [TO COMPLETE: Replace this approximate runtime with the measured runtime of the final run, and report the cache hit count or measured speed-up if available.]

Chapter 5

Experimental Methodology

5.1 Simulation and Training Configuration

The reported experiments used a population of 30 and five generations. Each locomotion rollout was configured for seven seconds. The magnetic-field sweep used 0.001, 0.008 and 0.05 T, and the fresh-run random seed was 42. The comparison figures therefore describe a short preliminary experiment rather than a full convergence study. Only one seed is documented in the available run configuration, so variation between independent runs has not yet been measured.

The same simulation and objective definitions were used for attractive and repulsive response runs. A candidate had to pass the graph and collision checks before being evaluated; simulation instability was still recorded as a failed evaluation. **[TO COMPLETE: Confirm the exact number of completed generations for each reported output directory and add the final training hyperparameters if they differ from the current source configuration.]**

5.2 Baseline Comparison Arm

The random baseline uses the same genotype, grammar masks, collision gate, simulation, objective calculation and NSGA-III survival procedure as the RL-assisted arm. Its only change is that action types and parameters are selected uniformly rather than by the trained PPO policy. This makes the comparison useful because any difference is more directly associated with learned variation choices, rather than with a different

simulator or selection rule.

5.3 Evaluation Metrics

Hypervolume (HV) is the main many-objective convergence measure because it combines improvement and spread of the non-dominated set. GD, IGD and IGD+ were not used because this engineering problem has no known true Pareto front; constructing a best-known front from the same experiments would make the comparison partly circular. Best and mean scalarised fitness are included as a simpler view of population progress, but they are not used as the NSGA-III selection criterion.

The action distribution records how often each structural operator is chosen and can reveal collapse of mutation or crossover actions. Per-action reward and policy-loss plots help investigate why an action becomes common or rare. Finally, collision rate, defined as $n_{\text{collided}}/n_{\text{offspring}}$, indicates how many proposed offspring fail the geometric feasibility check. These metrics together describe optimisation progress and the behaviour of the operator policy.

Chapter 6

Results and Discussion

6.1 Evolutionary Convergence and Hypervolume



Figure 6.1: Fitness trends: best and mean raw population fitness across generations, for each of the four objectives f_1 - f_4 .

Together, the scalarised fitness and hypervolume plots provide two views of progress. The scalarised curve gives a compact summary, whereas hypervolume considers improvement and spread across all four objectives. The plotted curves appear to improve over the early generations, with later changes that should be read from the final figure and reported numerically. This supports the implementation of the evolutionary

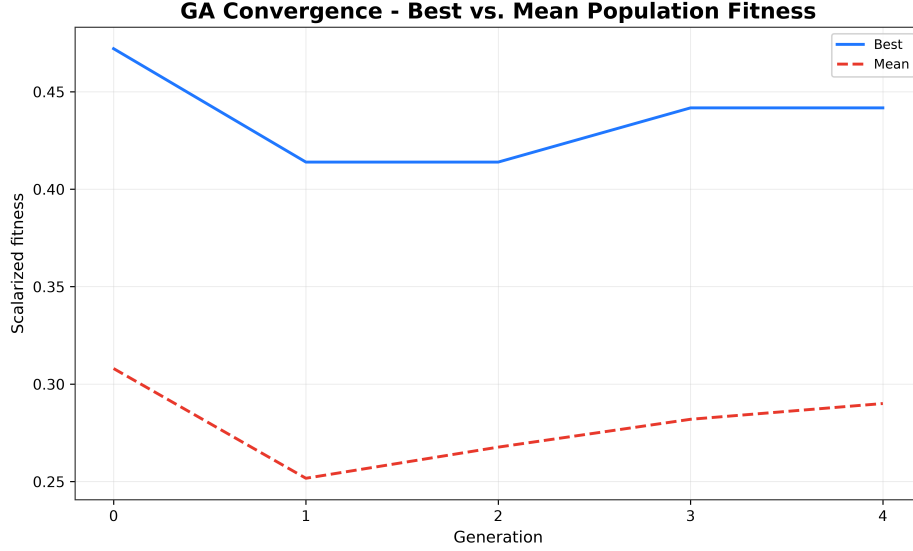


Figure 6.2: GA convergence: best and mean scalarized population fitness (objectives `api.scalarize`, equal-weight normalized combination of f_1 - f_4) across generations.

loop and gives partial evidence for O-4, but the short run is not sufficient to establish long-term convergence.

6.2 Pareto Front and Trade-off Analysis

The final population contains solutions with different compromises between speed, folding complexity and light response. A four-objective problem can produce many rank-zero individuals because one candidate may be better in one objective and worse in another. This dominance resistance is expected for a modest population and should not be interpreted as proof that all designs are equally useful. [The visible spread and rank colours in the Pareto figure should be checked against the final plot before adding numerical claims about particular trade-offs.](#)

6.3 Morphological Complexity versus Locomotion

The entropy–velocity plot addresses AO-2 by comparing a measure of change during folding with the resulting gait speed. [The plotted point cloud should be used to state whether high entropy gain is associated with lower, higher or similarly varied velocity, and whether later-generation points occupy a different region.](#) A visible concentration

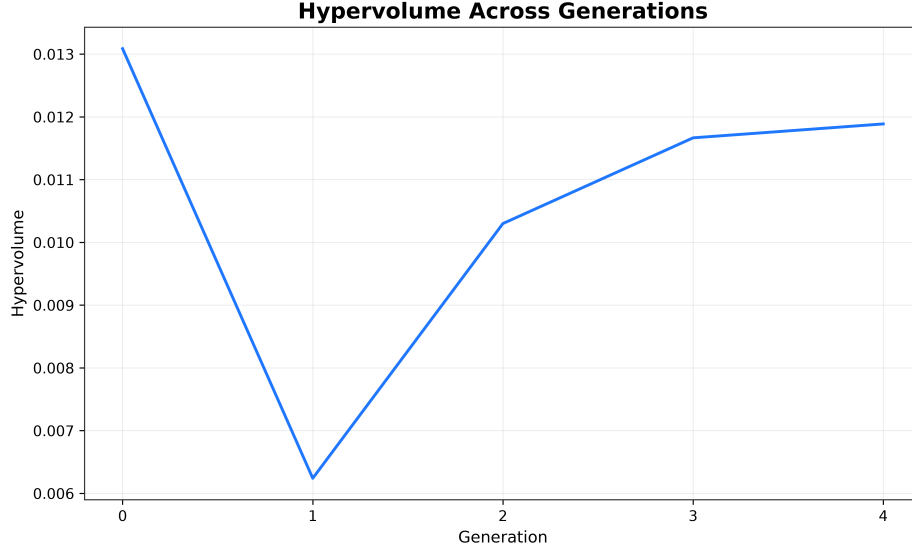


Figure 6.3: Hypervolume of the non-dominated front, per generation, normalized against the run-wide observed objective range (Section 5.3). Rising hypervolume indicates the evolving population’s Pareto front is expanding/improving in aggregate, independent of any single scalarization choice.

or slope would suggest that folding complexity is not independent of locomotion, while a broad cloud would suggest that connectivity and actuation dominate this relationship. The available evidence supports examining the relationship, but it does not justify a causal claim.

6.4 Evolved Morphologies: Attractive and Repulsive Response

The morphology figures provide end-to-end evidence for O-1 to O-3: graph designs were generated, converted into assemblies, folded and evaluated in the physics simulator. They also show how AO-1 was changed from surrogate fitness prediction to light-response objectives. [The images appear to show bilateral structures with light-sensitive joints marked in green; the module counts, joint counts and any consistent attractive-versus-repulsive difference should be confirmed directly from the rendered individuals.](#) The two response modes should therefore be read as separate design problems, not as proof that one morphology can optimise both behaviours at once.

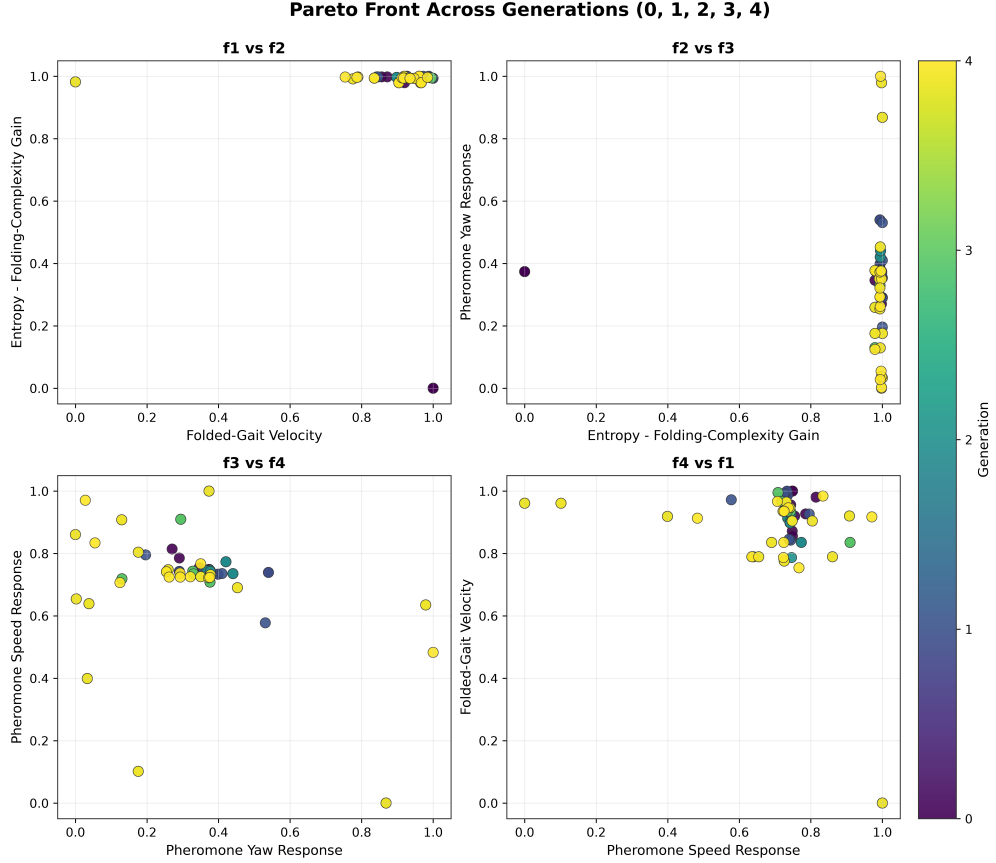


Figure 6.4: Pairwise Pareto front of the final generation’s population across all four objective pairs (f_1 vs f_2 , f_2 vs f_3 , f_3 vs f_4 , f_4 vs f_1), each axis min-max normalized within the generation. Points are coloured by NSGA-III non-dominated rank.

6.5 RL-Assisted versus Random-Baseline Comparison

At the scale shown, the random baseline has a higher best scalarised fitness than the RL-assisted arm. This is an important negative result: the learned policy has not yet demonstrated an advantage over uniform random variation. The comparison covers only five generations and a population of 30, so it should be treated as preliminary. A stronger conclusion would require longer runs and several independent seeds under the same settings.

The collision-rate plot should be used to determine whether the RL policy reduced failed proposals even while losing on fitness. [The relative collision-rate trend must be read from the figure before describing one arm as better or worse; the current text deliberately avoids inventing a direction or value.](#) The action-distribution result gives a plausible explanation for the fitness gap: crossover became rare, limiting the policy’s

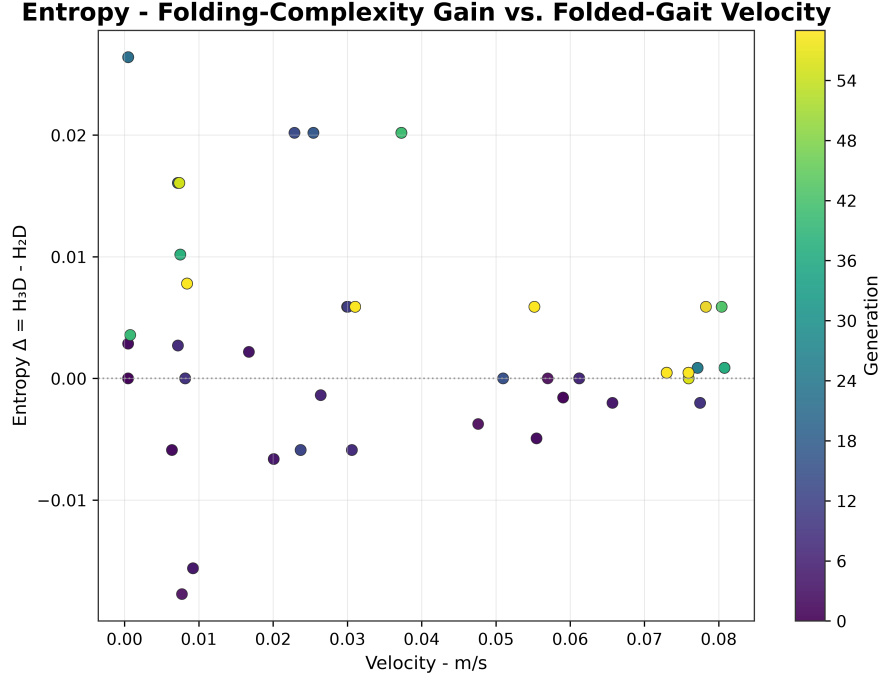


Figure 6.5: Folding-complexity gain ($f_2 = \tilde{H}_{3D} - \tilde{H}_{2D}$) versus folded-gait velocity (f_1), every individual across every generation, coloured by generation.

access to potentially useful structural combinations.

6.6 RL Training Behaviour and Diagnostics

The action-distribution diagnostic indicates that crossover actions became uncommon during training. The available implementation notes associate this with early entropy reduction and a reward-scale problem in which collision penalties could dominate ordinary fitness improvements. This interpretation is consistent with the baseline result, but it should be checked with the stored action counts and reward values rather than inferred from the plots alone.

The diagnostic figures separate policy behaviour from evolutionary selection. [The direction and magnitude of changes in reward, entropy coefficient, policy loss and value loss should be transcribed from the final plots if exact training claims are required.](#) At the time of writing, the redesign and its effect on crossover use have not been validated by a full-scale comparison. It is therefore more accurate to describe this section as diagnosis and partial investigation than as proof of successful RL improvement.

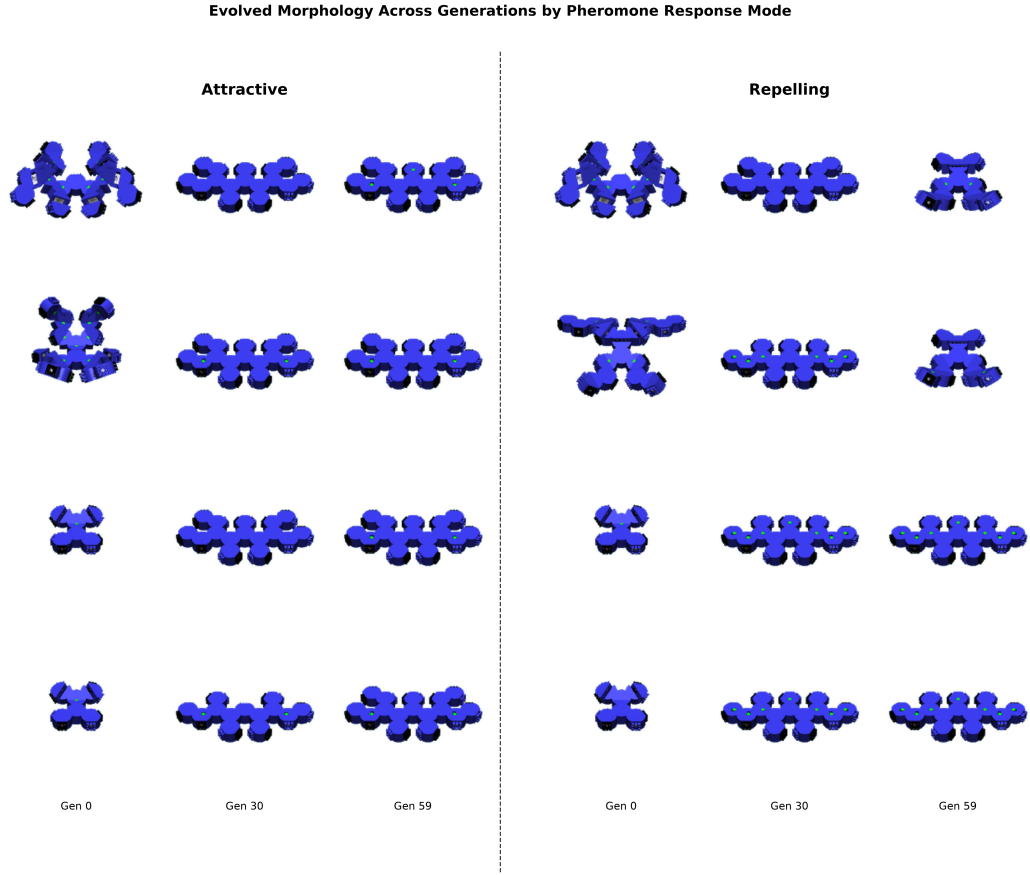
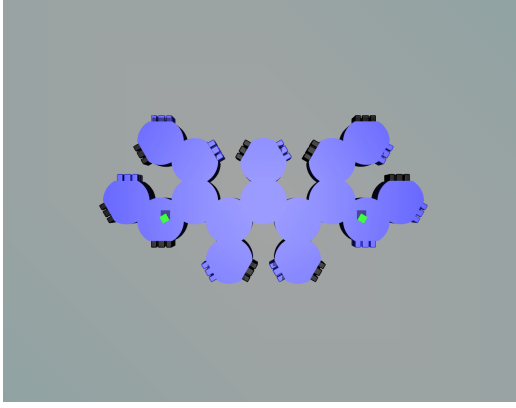


Figure 6.6: Representative evolved morphologies sampled across generations for both pheromone-response configurations (attractive, repulsive), rendered from the MuJoCo digital twin at a fixed shared scale so relative morphology size is directly comparable across the grid.

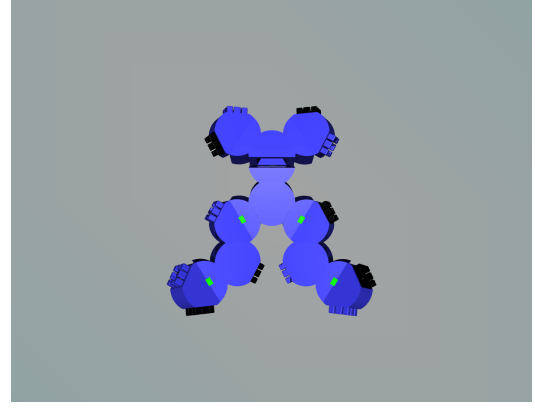
6.7 Discussion

O-1 was partially achieved. The graph policy and its integration were implemented, but the random baseline achieved the higher best scalarised fitness in the short comparison (Fig. 6.8). O-2 and O-3 were achieved at simulation level: legal graphs were folded, actuated and scored in MuJoCo, as illustrated by the evolved assemblies in Fig. 6.7. O-4 was addressed through the convergence, hypervolume and Pareto analyses (Figs. 6.2, 6.3 and 6.4), although the short run limits the strength of the conclusion.

AO-1 was changed and partially achieved through the attractive and repulsive pheromone-response objectives and their morphology results (Fig. 6.7). AO-2 was addressed by comparing entropy gain with folded-gait velocity (Fig. 6.5); this relationship should not be treated as statistically established without longer, multi-seed experiments. The



(a) Attractive configuration



(b) Repulsive configuration

Figure 6.7: Representative individual evolved morphologies, attractive (a) versus repulsive (b) pheromone-response configuration, both rendered from the MuJoCo digital twin. Green markers indicate light-sensitive joints.

overall system is functional, while the advantage of the learned variation policy remains unresolved.

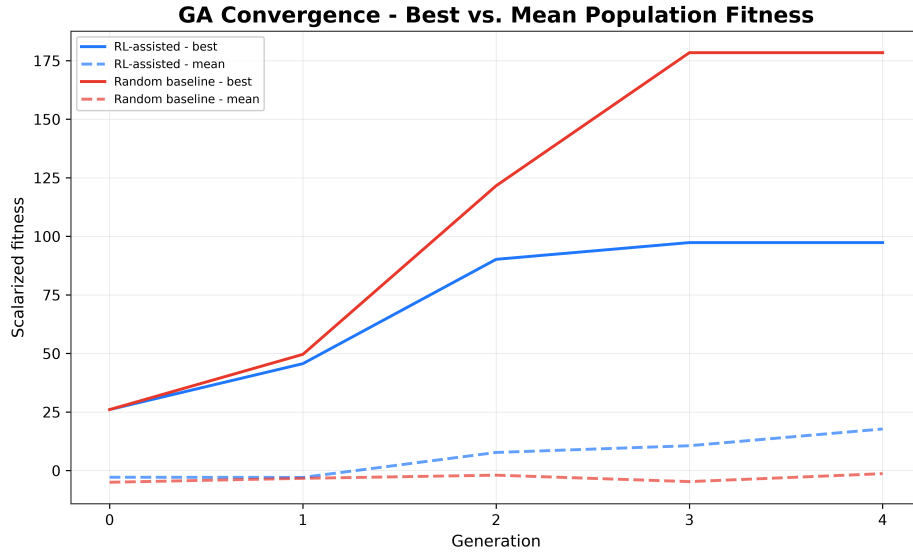


Figure 6.8: Best and mean scalarized fitness, RL-assisted breeding versus the uniform-random baseline (Section 5.2), over a preliminary 5-generation comparison run.

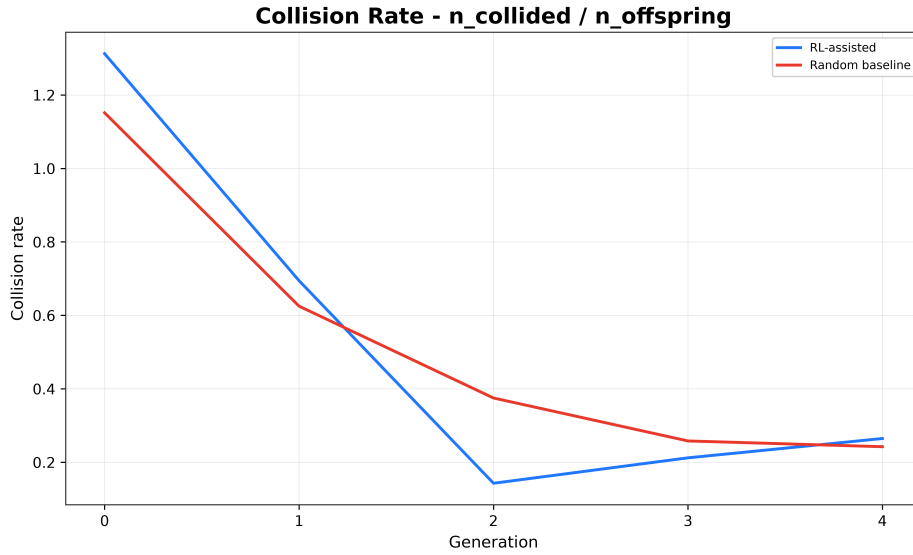


Figure 6.9: Collision rate ($n_{\text{collided}}/n_{\text{offspring}}$), RL-assisted versus random-baseline breeding, over the same preliminary comparison run.

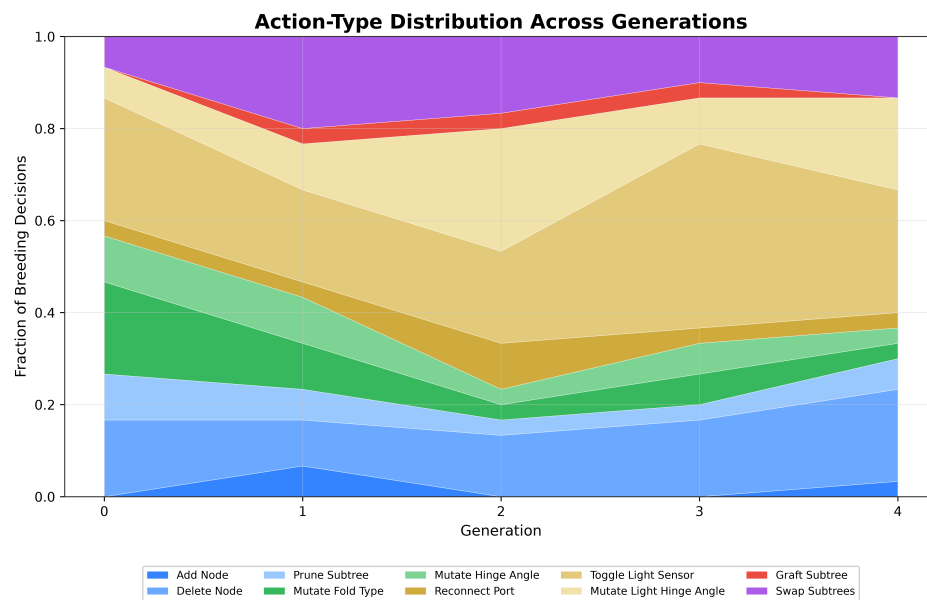
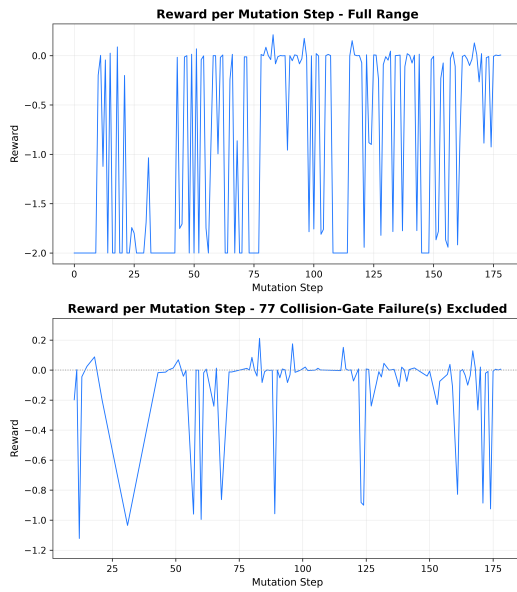
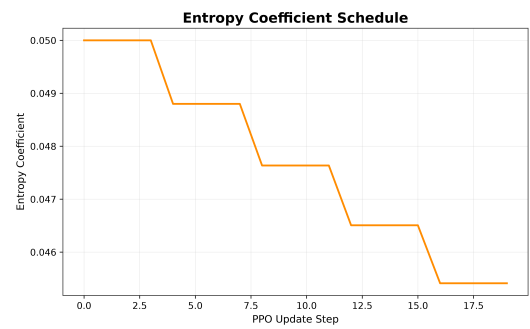


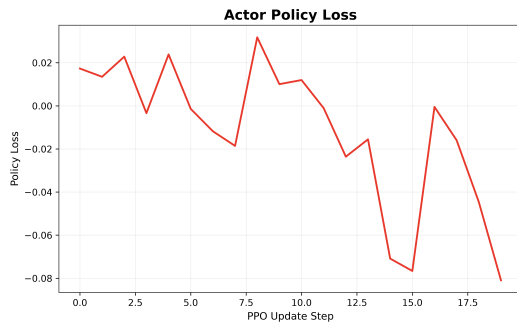
Figure 6.10: Fraction of breeding decisions by grammar action type, per generation.



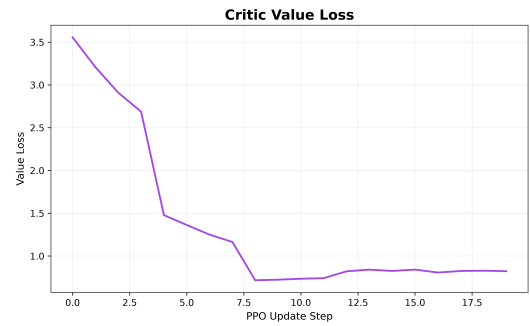
(a) Reward per mutation step



(b) Entropy coefficient schedule



(c) Actor policy loss



(d) Critic value loss

Figure 6.11: PPO training diagnostics across the run.

Chapter 7

Conclusions and Future Work

7.1 Summary of Contributions

The project delivered a graph representation, grammar masks and symmetry construction for Roblet morphologies. These components supported the valid evolved assemblies shown in Fig. 6.7. A MuJoCo digital twin was connected to magnetic actuation and objective evaluation, providing the end-to-end simulation evidence for O-2 and O-3.

The four-objective formulation and NSGA-III loop produced convergence, hypervolume and Pareto outputs (Figs. 6.2, 6.3 and 6.4). The light-response objectives provided the revised form of AO-1, while the entropy–velocity analysis addressed AO-2 (Fig. 6.5). Finally, the PPO operator policy, random baseline and diagnostics identified the unresolved crossover-use problem rather than hiding it (Figs. 6.8 and 6.10).

7.2 Limitations

The main limitation is scale. The documented configuration contains only five generations and one seed, so the convergence and RL-versus-random results are preliminary. The random-baseline result cannot establish a general advantage without longer runs and repeated seeds.

The light-response tests were performed on one simulated robot at a time. They do not demonstrate communication or coordination in a multi-robot swarm. The evolved morphologies were also not tested on physical Roblet hardware, so the conclusions

are limited by the accuracy of the digital twin and its contact, folding and magnetic assumptions.

The RL diagnostics identified a remaining issue: crossover actions became uncommon, and the available redesign has not yet been validated at full scale. [TO COMPLETE: Add measured crossover proportions, collision rates and before/after diagnostic values if available in the final run files.]

7.3 Future Work

The next stage should run at least 300 generations with a larger population and a reference-direction count chosen for that population size. This would test whether the early trends persist and whether the Pareto set becomes more informative as more designs are evaluated.

The RL and random arms should then be repeated under identical settings with five independent seeds. This would provide a practical first estimate of run-to-run variation without treating one favourable run as a general result. The crossover collapse should be re-evaluated after reward rescaling and entropy-schedule changes, with action proportions reported explicitly.

The best evolved designs should be checked on real Roblet units. This would test the assumptions about folding, connector placement, magnetic torque and light-sensitive joints. Further useful studies include ablations of the shared encoder, sensitivity to the entropy schedule and tests of reward-normalisation warm-starting. These experiments would show whether the learned policy contributes beyond random variation and whether the simulated light responses transfer to hardware.

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Appendix A

Glossary of Abbreviations

EA	Evolutionary algorithm
GNN	Graph neural network
HV	Hypervolume
MJCF	MuJoCo model format
MOO	Multi-objective optimisation
NSGA-III	Non-dominated Sorting Genetic Algorithm III
PPO	Proximal policy optimisation
RL	Reinforcement learning

Appendix B

Extended Results Figures

The extended figures and diagnostic files are retained in the project output directories.

[TO COMPLETE: Insert the names or paths of any additional figures included in the submitted appendix.]

Appendix C

Repository and Code Structure

The repository is organised into three main areas. The `src/` directory contains the simulation, genotype grammar, objective functions, reinforcement-learning policy, multi-objective loop and plotting code. The `helper_scripts/` directory contains MJCF generation and result-visualisation utilities. The `paper/` directory contains the paper draft, hyperparameter material and figure-generation resources. Input graph descriptions are stored in `graphs/`, generated models in `models/`, and run outputs in the `output/` and `old_output/` directories. **[TO COMPLETE: Add the approved repository link or access instructions if the code will be made available to the examiner.]**